A Self-Validating 12-Digit Identifier System

Complete Specification: Structure, Classification Tables, Parity Logic & Checksum

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1. Identifier Structure

Every identifier in this system is exactly 12 digits: $D_1D_2D_3D_4D_5D_6D_7D_8D_9D_{10}D_{11}D_{12}$

```
┌──────────┬──────────────┬──────────┬──────────┐
│ D₁-D₆    │ D₇-D₈        │ D₉-D₁₁   │ D₁₂      │
│ Date     │ Region Code  │ Sequence │ Check Digit │
│ YYMMDD   │ 2-digit index │ 000-999  │ Computed │
└──────────┴──────────────┴──────────┴──────────┘
```

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2. Field Breakdown — Example: 900523075049

```
9 0 | 0 5 | 2 3 | 0 7 | 5 0 4 | 9
 YY   MM    DD    RC    SEQ    CHK
```

Field	Digits	Value	Meaning
Year (YY)	D_1D_2	90	Year component (last 2 digits)
Month (MM)	D_3D_4	05	5th month
Day (DD)	D_5D_6	23	23rd day
Region Code (RC)	D_7D_8	07	Region index 07 (see lookup table)
Sequence (SEQ)	$D_9D_{10}D_{11}$	504	504th assignment for this date+region
Check Digit (CHK)	D_{12}	9	Computed checksum + parity classifier

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3. Century Disambiguation

The year field YY only provides two digits. The century is determined by the first digit of the sequence field (D_9):

D_9 Value	Century	Full Year Formula	Example
0, 1, 2, 3, 4	2000s	2000 + YY	YY=05, D_9=0 → 2005
5, 6, 7, 8, 9	1900s	1900 + YY	YY=90, D_9=5 → 1990

For our example: $D_9 = 5 \geq 5$ → 1900s → Full year = 1900 + 90 = **1990**

Complete date: **1990-05-23**

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4. Region Code Lookup Table (D_7–D_8)

4.1 Primary Codes (01–16)

Code	Region
01	Region Alpha
02	Region Bravo
03	Region Charlie
04	Region Delta
05	Region Echo
06	Region Foxtrot
07	Region Golf
08	Region Hotel
09	Region India
10	Region Juliet
11	Region Kilo
12	Region Lima
13	Region Mike
14	Region November
15	Region Oscar
16	Region Papa

4.2 Legacy / Alternate Codes (21–59)

Some regions have multiple codes due to historical sub-division:

Code Range	Maps To
21–24	Region Alpha
25–27	Region Bravo
28–29	Region Charlie
30	Region Delta
31, 59	Region Echo
32–33	Region Foxtrot
34–35	Region Golf
36–39	Region Hotel
40	Region India
41–44	Region Juliet
45–46	Region Kilo
47–49	Region Lima
50–53	Region Mike

Code Range	Maps To
54–57	Region November
58	Region Oscar

4.3 External / Foreign Codes (60–99)

Code	Classification
60	External Zone A
61	External Zone B
62	External Zone C
63	External Zone D
64	External Zone E
65	External Zone F
66	External Zone G
67	External Zone H
68	External Zone I
71–72	Legacy external (pre-2001 assignments)
74	External Zone J
75	External Zone K
76	External Zone L
77	External Zone M
78	External Zone N
79	External Zone O
82	Unclassified
83	Asia-Pacific group
84	South American group
85	African group
86	European group
87	British-Irish group
88	Middle Eastern group
89	Far Eastern group
90	Caribbean group
91	North American group
92	Eastern European / Post-Soviet group
93	Miscellaneous / Other
98	Unaffiliated entity
99	Unspecified / Special case

For our example: $D_7D_8 = 07 \rightarrow$ **Region Golf**

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5. Parity Classification (D_{12})

The check digit simultaneously serves as a binary classifier:

D_{12} Parity	Classification	Digit Values
Odd	**Class A**	1, 3, 5, 7, 9
Even	**Class B**	0, 2, 4, 6, 8

The assigning system does not freely choose D_{12}. It iterates through sequences ($D_9D_{10}D_{11}$) until the computed check digit's parity matches the desired classification.

For our example: $D_{12} = 9 \rightarrow$ Odd $\rightarrow$ **Class A**

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6. Sequence Assignment Logic

The system assigns identifiers using this algorithm:

```
INPUT:  date (YYMMDD), region (RC), desired class (A or B)

FOR seq = 000 to 999:
    candidate = YYMMDD + RC + seq (11 digits)
    C = compute_checksum(candidate)

    IF C = 10:
        SKIP (no single-digit representation)

    ELSE IF C is odd AND desired class is A:
        ASSIGN → identifier = candidate + C
        STOP

    ELSE IF C is even AND desired class is B:
        ASSIGN → identifier = candidate + C
        STOP

    ELSE:
        SKIP (parity mismatch)
```

What this means in practice:

- ~9.1% of sequences are impossible (C = 10)
- ~45.5% of sequences yield odd check digits (Class A)
- ~45.5% of sequences yield even check digits (Class B)
- The system never "picks" the last digit — it is always computed

7. Checksum Algorithm (ISO 7064 Mod 11,2)

7.1 Weight Function

```
W(i) = 2^(i-1) mod 11    for i = 2, 3, ..., 12
```

Constant values:

```
W(2)=2  W(3)=4  W(4)=8  W(5)=5  W(6)=10  W(7)=9
W(8)=7  W(9)=3  W(10)=6  W(11)=1  W(12)=2
```

7.2 Formula

Given: $S = D_1D_2D_3D_4D_5D_6D_7D_8D_9D_{10}D_{11}$
Let: $R = \text{reverse}(S)$

$$\Sigma = \Sigma_{j=0}^{10} R[j] \times W(j+2)$$

$$C = (12 - (\Sigma \bmod 11)) \bmod 11$$

Valid iff: $C = D_{12} \;\wedge\; C \in \{0,1,\ldots,9\}$

7.3 Edge Cases

Condition	Result
$C \in \{0–9\}$	Valid check digit, assign normally
$C = 10$	Impossible sequence, skip entirely
(12 - remainder) = 11	Wraps to 0 via mod 11
(12 - remainder) = 12	Wraps to 1 via mod 11

8. Worked Example — Full Computation

Identifier: `900523075049`

Step 1 — Extract first 11 digits and reverse:

```
S = [9, 0, 0, 5, 2, 3, 0, 7, 5, 0, 4]
R = [4, 0, 5, 7, 0, 3, 2, 5, 0, 0, 9]
```

Step 2 — Multiply each reversed digit by its weight:

```
R[0]  × W(2)  =  4 × 2  = 8
R[1]  × W(3)  =  0 × 4  = 0
R[2]  × W(4)  =  5 × 8  = 40
R[3]  × W(5)  =  7 × 5  = 35
R[4]  × W(6)  =  0 × 10 = 0
R[5]  × W(7)  =  3 × 9  = 27
R[6]  × W(8)  =  2 × 7  = 14
R[7]  × W(9)  =  5 × 3  = 15
```

```
R[8]  × W(10) =  0 × 6  = 0
R[9]  × W(11) =  0 × 1  = 0
R[10] × W(12) =  9 × 2  = 18
```

Step 3 — Sum all products:

```
Σ = 8 + 0 + 40 + 35 + 0 + 27 + 14 + 15 + 0 + 0 + 18 = 157
```

Step 4 — Modular reduction:

```
Σ mod 11 = 157 mod 11 = 3

(Verification: 14 × 11 = 154,  157 - 154 = 3)
```

Step 5 — Compute check digit:

```
C = (12 - 3) mod 11
C = 9 mod 11
C = 9
```

Step 6 — Validate:

```
Computed:  C    = 9
Actual:    D₁₂  = 9
Match:     9 = 9  ✓

Parity:    9 is odd → Class A  ✓
```

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9. Complete Decoded Summary

```
Identifier:      900523075049
```

90	05	23	07	504	9
YY	MM	DD	RC	SEQ	CHK

```
Date:             1990-05-23  (D₉=5 ≥ 5 → 1900s)
Region:           07 → Region Golf
Sequence:         504
Check Digit:      9 (computed, verified ✓)
Classification:   Class A (odd)
Checksum Valid:   YES
```

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10. System Properties

Property	Description
Error detection	Any single-digit error in D_1–D_{11} causes checksum mismatch
Dual-purpose D_{12}	Simultaneously encodes binary classification and integrity check

Property	Description
Impossible sequences	~9.1% of sequences yield C=10, are permanently excluded
Parity-balanced	~50/50 split between Class A and Class B assignments
Base	Modular arithmetic over Z_{11}
Standard	ISO 7064 Mod 11,2
Self-validating	Any recipient can verify integrity with just the 12 digits — no external lookup required

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This is a complete, self-contained specification for a structured numeric identifier system with embedded date encoding, regional classification, binary parity assignment, and weighted modular checksum validation.